Can your model A stop expanding and still fit? A bound-expansion fit

Addendum to the integrated model at $R = 14.3$ Gpc

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1 Constraints

We use the integrated fixed-profile model of the previous addendum, with its geometry, $e^{-r/L}$ profile, sine term and start at $0.01\,t_0$. The CoE mass and an optional uniform background of mass M_b act on every particle through

$$\ddot{\mathbf{x}} = -\frac{G\,[M(<r) + M_b(<r)]}{r^3}\,\mathbf{x}, \qquad M_b(<r) = M_b\,\min\bigl(1,\, r^3/D^3\bigr)\,, \tag{1}$$

and Earth's specific energy today uses the exact potential of both,

$$E = \tfrac{1}{2}v_E^2 + \Phi(r_E), \qquad \Phi(r) = -\frac{GM_{\rm tot}}{L}\left[\frac{f(u)}{u} + \frac{(1+u)e^{-u}}{2}\right] + \Phi_b(r), \qquad u = \frac{r}{L}, \quad M_{\rm tot} = \frac{M}{f(D/L)}, \tag{2}$$

with $f(u) = 1 - e^{-u}(1 + u + u^2/2)$ and $\Phi_b = -GM_b(3D^2 - r^2)/2D^3$ inside D, $-GM_b/r$ outside. A fit is accepted only if $E \le 0$, so that Earth eventually turns around, and if every particle's radial velocity stays positive up to today. We quote the bound fraction as $(v_E/v_{\rm esc})^2 = v_E^2/[-2\Phi(r_E)]$, which is at most 1 for a bound Earth. The turnaround time comes from continuing Earth's radial motion past today. The background is a uniform sphere of radius D centred on the CoE and fixed in space. We did not let it expand with the flow, because comoving uniform matter in a flow with $H = 1/t$ has a density parameter that grows as t_0/t toward early times. At $\Omega_b = 0.3$ that is 30 at the start, so everything recollapses long before today, which is the same ill-posedness that ruled out the comoving CoE profile. Truncating it at D adds its mass to the mass inside D and keeps its potential finite.

Migkas plot $A/A_{\rm all}$, the ratio to the all-sky mean. Your sheet instead divides by the unperturbed kR, which is harmless when gravity is weak but not when it is strong enough to bind. We therefore fit both forms,

$$A_{\rm literal}(\theta) = [1 - {\rm Amp}\sin(\theta + \varphi)]\,\frac{|\Delta\dot{\mathbf{x}}|}{kR}, \qquad A_{\rm sky}(\theta) = [1 - {\rm Amp}\sin(\theta + \varphi)]\,\frac{|\Delta\dot{\mathbf{x}}|}{\langle|\Delta\dot{\mathbf{x}}|\rangle_{\rm sky}}. \tag{3}$$

Only the combination $\mu = GMt_0^2/D^3$ enters the dynamics, so the age and the mass are degenerate. With $H_0 = 70$ $\mathrm{km\,s^{-1}\,Mpc^{-1}}$ fixed as Earth's expansion rate today, each solution sets its own age and mass,

$$t_0 = \frac{(Ht)_E}{H_0}, \qquad (Ht)_E = \frac{v_E\,t_0}{r_E}, \qquad M = \frac{\mu D^3}{Gt_0^2}, \qquad \frac{M}{M_{\rm crit}} = \frac{2\mu}{(Ht)_E^2}. \tag{4}$$

2 Results

With your literal normalisation, no bound solution fits. Binding Earth requires deceleration of order the expansion itself, and that lowers every relative velocity together. The mean of your model A then drops well below the Migkas mean of 1, which ruins R^2 (dotted curve in Fig. 1). The only way to be bound without that deceleration is to spread the mass far beyond D ($L/D \approx 2.5$), so that the potential is deep while the tidal pull on the shell is weak. That leaves almost no anisotropy, and R^2 falls to the level of the sine term alone (Table 1). Adding the uniform background makes this worse.

Normalised to the sky mean, as Migkas's $A/A_{\rm all}$ is, the uniform part of the deceleration drops out. Bound solutions then fit as well as the unbound one, and slightly better (Table 1). In fact the unconstrained sky-normalised optimum is already bound, so constraint (a) costs nothing. These solutions have about one critical mass inside D with $L/D \approx 0.52$–0.64 and $D/R \approx 1.4$–1.6, and Earth turns around at 2.8–3.8 t_0. A background of $\Omega_b = 0.3$ today takes over part of the binding, lowers the CoE mass to 0.63–0.83 critical, delays turnaround to 3.7–5.6 t_0 and costs less than 0.01 in R^2. Freeing Ω_b with ℓ_0 also finds a branch with a large background and almost no CoE mass. Its anisotropy comes from the artificial edge of the background at $r = D$, so we do not regard it as physical.

The age is the tighter limit. Under attraction alone the expansion only slows, so $(Ht)_E < 1$ always and $t_0 < 1/H_0 = 14.0$ Gyr. That also rules out any age of order $r/c = 46.6$ Gyr without a repulsive term. The unrestricted bound optima have $(Ht)_E \approx 0.79$–0.83, so they are 11.1–11.6 Gyr old and fall below the 12 Gyr oldest-star limit. They are flagged in Table 1. Requiring an age of at least 12 Gyr as well, which means $(Ht)_E \ge 0.859$, costs almost

nothing. The fit moves to a lighter, more extended mass of about 0.66 critical ($L/D \approx 0.44$) that is still bound, keeps $R^2 = 0.898$ with ℓ_0, and turns around after about 8 ages, near 100 Gyr. With $\Omega_b \approx 0.3$ the same holds at 0.44 critical, with turnaround after about 20 ages. Asking for 13 Gyr is too much: the best solution then sits exactly on the bound edge ($E = 0$, turnaround only at infinity), and R^2 falls to 0.72. A bound universe with $H_0 = 70$ that fits the dipole can therefore just meet the oldest stars at 12 Gyr, but not 13 Gyr. This is the classic age problem of a decelerating universe.

Table 1: Constrained best fits, vs the Migkas points unless marked. All rows keep every particle moving outward today; "bound" adds $E \leq 0$ for Earth. $M/M_{\rm crit}$ and Ω_b are fractions of the critical mass inside D at $H_0 = 70$ today. Age is t_0 from Eq. (4), with a dagger for ages below 12 Gyr; the last block adds the age limit to the bound fit; turnaround is in units of that age. Rows with $(v/v_{\rm esc})^2 = 1$ sit on the bound edge and turn around only at infinity. The sky-normalised fits without the bound constraint are identical to the bound rows.

Case	D/R	L/D	$M/M_{\rm crit}$	Ω_b	ℓ_0 [°]	Amp	φ	$R^2_{\rm Migkas}$	$R^2_{A_F}$	$(v/v_{\rm esc})^2$	Age [Gyr]	Turnaround [t_0]
Literal, unbound allowed	2.71	0.072	0.0248	0.00	0.0	−0.052	0.51	0.755	0.829	34.5	13.7	∞
Literal, unbound allowed, ℓ_0	2.90	0.070	0.0273	0.00	13.2	−0.051	0.75	0.894	0.979	30.7	13.6	∞
Literal, bound	5.50	2.633	0.0367	0.00	0.0	−0.052	0.57	0.277	0.307	1	13.9	∞
Literal, bound, ℓ_0	5.45	2.428	0.0423	0.00	14.0	−0.052	0.82	0.279	0.309	1	13.9	∞
Literal, bound, $\Omega_b = 0.3$	1.05	3.000	0.0187	0.31	0.0	−0.047	6.18	−0.004	−0.008	1	13.3	∞
Sky-normalised, bound	1.37	0.641	1.31	0.00	0.0	−0.116	5.13	0.776	0.853	0.188	11.1†	2.80
Sky-normalised, bound, ℓ_0	1.56	0.566	1.06	0.00	12.4	−0.086	5.17	0.899	0.984	0.274	11.4†	3.39
Sky-normalised, bound, vs A_F	1.36	0.570	1.12	0.00	0.0	−0.116	5.13	0.776	0.853	0.255	11.3†	3.23
Sky-normalised, bound, vs A_F, ℓ_0	1.52	0.524	0.963	0.00	12.3	−0.088	5.16	0.899	0.985	0.331	11.5†	3.83
$\Omega_b = 0.3$, bound	1.34	0.490	0.832	0.30	0.0	−0.120	5.12	0.772	0.852	0.348	11.4†	3.70
$\Omega_b = 0.3$, bound, ℓ_0	1.50	0.442	0.701	0.30	12.6	−0.091	5.14	0.895	0.983	0.444	11.5†	4.63
$\Omega_b = 0.3$, bound, vs A_F	1.33	0.435	0.709	0.30	0.0	−0.119	5.12	0.771	0.852	0.443	11.5†	4.60
$\Omega_b = 0.3$, bound, vs A_F, ℓ_0	1.48	0.408	0.631	0.30	12.3	−0.092	5.14	0.895	0.983	0.516	11.6†	5.60
Ω_b free, bound, ℓ_0 (edge branch)	2.19	0.068	0.00277	5.31	192.1	−0.150	4.46	0.903	0.971	0.113	8.7†	1.54
Age $\geq$ 12 Gyr	1.45	0.436	0.664	0.00	0.0	−0.090	5.26	0.773	0.850	0.604	12.0	8.32
Age $\geq$ 12 Gyr, ℓ_0	1.66	0.436	0.664	0.00	12.5	−0.067	5.31	0.898	0.983	0.604	12.0	8.33
Age $\geq$ 12 Gyr, vs A_F, ℓ_0	1.59	0.421	0.652	0.00	12.4	−0.072	5.27	0.898	0.984	0.634	12.0	9.30
$\Omega_b \approx 0.3$, age $\geq$ 12 Gyr	1.42	0.351	0.444	0.28	0.0	−0.094	5.24	0.768	0.849	0.781	12.0	18.55
$\Omega_b \approx 0.3$, age $\geq$ 12 Gyr, ℓ_0	1.60	0.346	0.44	0.28	12.7	−0.071	5.27	0.894	0.982	0.792	12.0	20.12
$\Omega_b \approx 0.3$, age $\geq$ 12 Gyr, vs A_F, ℓ_0	1.54	0.333	0.43	0.28	12.4	−0.076	5.24	0.893	0.982	0.82	12.0	24.85
Age $\geq$ 13 Gyr, ℓ_0	4.03	0.584	0.332	0.00	14.4	−0.042	0.56	0.719	0.789	1	13.0	∞

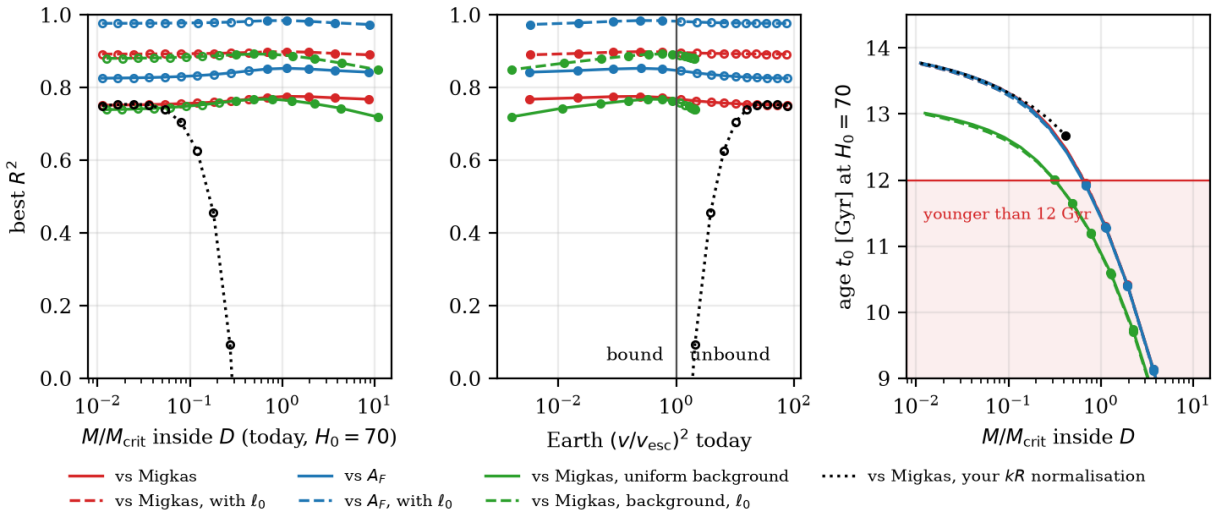


Figure 1: Best R^2 with D/R, L/D, Amp and φ (and ℓ_0 where marked) re-fitted at each CoE mass, sky-normalised except the dotted curve. Filled points have a bound Earth. The green curves add a uniform background of 0.3 critical at the unperturbed rate, which is about 0.45 at today's rate. The right panel gives the age at $H_0 = 70$; the shaded band is younger than the oldest stars. Curves end where particles turn around before today.